Department of Computer Science

Community-engaged Learning (CeL) Course Syllabus

Course Name: Software Engineering	**Number:** CS3365	**Semester: Fall** 2026
Instructor: Dr. Maaz Amjad	**Office: EC** 211B	**Email:** maaz.amjad@ttu.edu
Office Hours: MON 10:00 AM – 12:00 PM, WED 4:00 PM – 5:00 PM, or by appointment (M-F)	**Location: EC** 211B	**Link:** See Canvas **Credit:** 3 Credits
TA/Grader:		**Email:**
TA/Grader Office Hours:		**TA/Grader Office:**

If you write me an email for this class, please start the email subject with [CS3365].

Catalogue Listing: Prerequisite: CS 2365, MATH 3342, or equivalent. Introduces theory and practice for software engineering. Topics include software life cycle, requirements, specification and analysis, software architecture and detailed design, implementation, and testing.

Textbooks:

1. Winters, T., Manshreck, T., & Wright, H. *Software Engineering at Google: Lessons learned from programming over time*. (1st Edition) O'Reilly Media, Inc, 2020. (*Recommended*) Free online edition: https://abseil.io/resources/swe-book
2. Ian Sommerville, *Software Engineering* (10th Edition), Pearson, 2016. (*Recommended*) (PDF will be provided on Canvas)
3. Bernd Bruegge and Allen Dutoit, *Object-Oriented Software Engineering Using UML, Patterns, and Java* (3rd Edition), Pearson, 2009. (*Recommended*) (PDF will be provided on Canvas)

Course Description: This course introduces students to the theory and practice of software engineering - focusing on systematic approaches to development, maintenance, and evolution of complex software systems.

Course Objectives:

1. Understanding basic principles of Software Engineering
2. Practice software engineering techniques
3. Learn modeling using UML modeling language
4. Secure program design

Writing Intensive:

CS 3365 is one of the two required writing intensive courses for the Department of Computer Science. According to the University, "the fundamental objective of a writing intensive course is for students to write often and receive critical review from the course instructor. Students should be required to rewrite, based on the instructor's critique. The writing intensive course emphasizes the process as well as the products of writing. Faculty use writing to reinforce student learning. Students' writing should formulate ideas, raise questions, and express considered opinions. Students' written work should analyze, integrate, and synthesize as well as communicate."

Key Topics:

1. Software Development Lifecycle
2. Requirements Elicitation
3. Requirements Analysis
4. Architectural Design
5. Design Patterns
6. Detailed Design
7. Introduction to Verification and Validation
8. Professionalism and Ethics
9. Secure program design
10. Local/global impacts of computing solutions on individuals, organizations, and society

Definition and Rationale for Community-engaged Learning

Community-engaged learning integrates academic study with sustained, reciprocal collaboration with external communities. In this course, the community partner is not simply a project client; the partner contributes expertise about the community context, helps define priorities and constraints, reviews student work, and provides students feedback on the usefulness and appropriateness of the proposed solution. This course follows hands-on experience, and students apply software engineering knowledge in a real-world setting while developing civic responsibility, professional judgment, communication skills, and an understanding of how computing decisions affect people and organizations. The intended result is mutual benefit: deeper student learning and a useful, responsible contribution to the partner.

Community-engaged Learning (CeL) Requirements

The CeL project is a required and integral component of this course (CS 3365). Students will work in instructor-approved teams (3-4 members in each team) and will participate in the complete software engineering process with the community partner. More details about a list of key Community partner organizations will be provided. Each team will meet at least 3 times with representatives of the partner organization, and complete the following engagement activities:

I. In this course, students will spend at least 30 hours in a community-engaged learning experience throughout the semester.

II. Students will be given a list of projects. The community partner organization (s) will be invited in partner orientation and share about their organization, project details, community context, intended users, existing workflow, constraints, and desired outcomes.

III. Students will meet stakeholders, identify requirements of projects, and ask questions to find community needs.

IV. Students will translate community needs into prioritized, testable, and traceable software requirements.

V. Students will co-design and iteratively refine a feasible solution through prototypes, sprint reviews, and partner feedback.

VI. Students will validate the solution with representative users or partner-designated reviewers and respond constructively to feedback.

VII. Students will deliver a final demonstration and an appropriate handoff package, which includes source code, technical documentation, user documentation, known limitations, and maintenance recommendations.

VIII. Students will complete individual reflection assignments that connect community experiences to software engineering concepts, ethics, civic responsibility, teamwork, and professional growth.

Roles and Responsibilities

A. **Community partner:** The community partner organization (s) will provide contextual expertise, help identify and prioritize needs, identify appropriate stakeholders, review project artifacts, communicate constraints, and evaluate the team's professionalism and responsiveness. The partner is not expected to serve as the team's technical supervisor.

B. **Students:** Prepare for and attend scheduled engagements; communicate professionally; document decisions; protect confidential or sensitive information; manage the project responsibly; contribute equitably; meet agreed deadlines; and produce work that is usable, maintainable, secure, and appropriate for the partner's context.

C. **Instructor and instructional team:** Approve scope and methods; teach and assess software engineering content; facilitate communication; monitor workload and risk; resolve project concerns; and make all final academic and grading decisions.

Professional, Ethical, Privacy, and Safety Expectations

1) Treat community members and partner representatives as collaborators and experts in their own context. Use respectful, inclusive, and accessible communication.
2) Do not collect, copy, upload, or use personally identifiable, confidential, regulated, proprietary, or production data unless the instructor and partner have explicitly approved the procedure in writing.
3) Use only private repositories and partner-approved collaboration platforms. Do not publish partner materials, code, screenshots, meeting recordings, or project details without written approval.
4) Do not promise features, deployment dates, costs, or long-term support on behalf of the course or university. Scope changes must be documented and approved by the course instructor.
5) Do not deploy software to a live production environment or conduct testing with real users without explicit instructor and partner approval.
6) Report communication problems, missed partner commitments, security concerns, ethical concerns, or team participation problems promptly to the instructor.

Course Prerequisite: CS 2365, MATH 3342, or equivalent

Expected Prior Knowledge and Skills: The successful student should have competent skills in procedural and object-oriented programming, knowledge of data structures and algorithm analysis, and knowledge of statistical and probabilistic mathematics.

***Note**: This course is one of the courses used as a primary measure of outcomes 4, 5, and 6.*

Learning Outcomes: Students who have completed this course should have

1. Be able to elicit and analyze customer requirements individually and with team members. (1, 5)
2. Be able to design software systems using modeling techniques individually and with team members. (2,5)
3. Understand verification and validation techniques. (2)
4. Professionalism and ethics. (4)
5. Understand the use of software engineering tools, templates and references. (6)
6. Be ability to use secure program design techniques. (6)
7. Understand local/global impacts of computing solutions on individuals, organizations, and society (4)

Learning Outcomes & Assessment Methods: Students who have completed this course should have the ability to:

Objective	ABET Outcomes (1-6)	Assessment Methods
Ability to elicit and analyze customer requirements individually and with team members.	1 5	Quizzes/Assignments
Ability to design software systems using modeling techniques individually and with team members.	2 5	Quizzes/Assignments/Projects
Understanding of verification and validation techniques.	2	Quizzes/Assignments
Professionalism and ethics.	4	Quizzes/Assignments
Understanding the use of software engineering tools, templates and references	6	Quizzes/Assignments/Projects
Ability to use secure program design techniques.	6	Quizzes/Assignments/Projects
Understanding local/global impacts of computing solutions on individuals, organizations, and society	4	Quizzes/Assignments

CeL Learning Outcomes, **Project Deliverables and Assessment Methods (**Part of Projects-based Learning: 50% of Total Course Credit, see below **"Grading Policy")**: Upon successful completion of the course, students should be able to demonstrate the following learning outcomes:

Student Learning Outcome	CeL Experiences / Assignments	Assessment Methods
Elicit, analyze, prioritize, validate, and document stakeholder requirements individually and with team members.	Partner orientation; stakeholder discovery; requirements review	Partner organization feedback based on requirements rubric
Design a software system using appropriate architectural, modeling, interface, data, and security techniques.	Architecture/design rationale; UML; prototype; design review	Design rubric; traceability; technical review
Implement, integrate, test, and evaluate a maintainable software solution using appropriate tools and engineering practices.	Private repository; sprint increments; code review; CI/testing	Repository evidence; code quality; test results; final product
Function effectively and accountably on a team, including planning work, resolving problems, and contributing equitably.	Team charter; sprint planning; meeting records; peer feedback; activity log	Peer evaluation; instructor observation; contribution evidence
Communicate technical information effectively to community members, users, and technical audiences.	Partner meetings; progress reports; demonstration; user and technical documentation	Writing and presentation rubrics; partner feedback
Recognize professional responsibilities and analyze ethical, privacy, security, accessibility, and societal impacts of the proposed solution.	Risk/ethics analysis; security and accessibility review; reflections	Ethical reasoning rubric; compliance checklist; reflection rubric
Use evidence, stakeholder feedback, and reflective practice to improve the solution and professional decision-making.	Prototype feedback; user validation; revisions; three reflections; final impact analysis	Iteration evidence; reflection quality; partner evaluation

Methods of Assessment of Learning Outcomes: The expected learning outcomes for the course will be assessed through quizzes (including quiz/pop quizzes), assignments (including in-class assignments), and projects.

Assignment Submission: All assignments must be submitted via TTU RaiderCanvas. No assignments should be submitted via email. Any assignment submitted via email will not be graded and will receive zero grade. In case you have any issues submitting assignments via TTU RaiderCanvas, please contact tech support and resolve the problem.

Partner-facing deadlines and scheduled partner meetings are professional commitments. Missed meetings, late communication, or late partner deliverables may reduce the CeL project and professionalism scores even when the general late-work window would otherwise apply.

Grading Policy: The final grade for this course will be based on the items described below:

- Quizzes **10%**
  - Quizzes are short tests given through Canvas or during class lecture (in-class) over the subject matter covered in the course.
  - For these quizzes, you **may** use a calculator and your notes, book, or other lecture materials.
- Assignments: **40%**
  - Assignments (group or individual) will be programming or written assignments to be done outside of class.
  - Occasionally, in-class assignments will be assigned with minimal or no warning (surprise assignments) which is due before the end of the class it was assigned. For in-class assignments, all the students should be physically present in the class.
  - Individual Reflection Assignments (expectations and timeline): Students will also complete four individual reflection assignments throughout the semester:
    I. **Reflection 1** (at the beginning of semester) **- Learning Goals and Expectations**: Describe your learning goals, expectations for the course, anticipated learning outcomes, and the knowledge and skills you hope to develop.
    II. **Reflection 2 (**before the CeL project) **- Community Context and Stakeholder Perspective:** Analyze the community partner's mission, the broader community context, stakeholder perspectives, relevant assumptions, and the relationship between the identified community need and the concepts covered in the course.
    III. **Reflection 3 (**during the CeL project) **- Midpoint Learning, Ethics, and Team Practice:** Evaluate the feedback your team have received, the effectiveness of your team's collaboration, ethical or professional challenges, relevant trade-offs, and specific changes needed in the project or in your own professional practices.
    IV. **Reflection 4 (**after the CeL project) **- Impact and Professional Learning Synthesis:** Assess the usefulness and limitations of the final contribution, provide evidence of learning and impact of CeL's project, and connect the experience to software engineering principles, professional responsibility, and civic engagement.
- Projects **50%**
  - There will be two Projects: Project 1: 25%, Project 2: 25%
  - Projects will be long programming or written assignments to be done outside of class. They may also consist of giving presentations to the instructor or the class.

Criteria for Grading: All assignments will be graded within two weeks of the due date unless unforeseen circumstances prevent timely grading. Table 1 shows the grading scale and breakdown of grade components.

Table 1: Scores are not rounded

letter grade	lower bound	upper bound
A	90%	100%
B	80%	89%
C	70%	79%
D	60%	69%
F	0%	59%

For example, if you receive an 89.9 score, it will be considered 89%, and you will still receive a B grade.

Additional Information

- The final exam will replace your lowest exam grade if the final exam score is higher than your lowest exam score. However, the final exam will not replace the grade on an exam you did not take.
- Deadlines will not be extended due to system failures or disk crashes. Please back up your files securely in the cloud (OneDrive is free for TTU students) to prevent the loss of work.
- Do not save your work to publicly available GIT repos – this increases your risk of being plagiarized.
  - This includes REPL.it or other online compilers that, by default, make your code publicly available.
  - Failure to abide by this rule will subject you to academic penalties even if your online code is used without your knowledge or permission.
  - Unless otherwise noted, assignments are due at 11:59 PM (End of Day).
- Late work will only be accepted within 72 hours of the due date/time with the following deductions unless prior arrangements have been made with the professor or otherwise stated in class:
  - 10% deduction for the first 24 hours
  - 25% deduction for the second 24 hours
  - 40% deduction for the third 24 hours
- Some assignments will not be accepted late and will be marked as such on Canvas.
- Quizzes will never be accepted late, regardless of reason.
- Students may not make up or submit any item once it has been graded and returned to any student.
- All questions concerning graded material must be submitted **in writing** along with the graded material by the last day of classes, as marked in the Academic Calendar.

COURSE SPECIFIC POLICIES

Attendance Policy

Regular and punctual attendance is mandatory for this course. As a student, you are required to attend all lecture sessions scheduled. These sessions are pivotal to your learning and provide crucial instruction on the course material. Furthermore, all announcements, assignments, and lecture materials covered in each session will be your responsibility, regardless of whether or not you were present in the lecture sessions.

1. **Responsibility for Class Material**: If you miss a lecture, you are still responsible for all course content covered, including any announcements made, assignments given, and material discussed during the lecture. It is recommended that you collaborate with fellow students to receive any missing lecture materials or learn of any announcements covered during the lecture. Contact the instructor by email or approach the instructor during office hours to discuss any possible make-up opportunities, assuming you are in accordance with the rest of this policy.
2. **Notification of Absence**: If you must be absent from class for any reason, you are required to notify the instructor in writing either prior to the absence or within a 48-hour window following the missed lecture. This notification should include the reason for your absence and a plan for making up any missed coursework. If you need to be absent for more than one week, please check these guidelines.

3. **Make-Up Work**: In cases of notified absences, arrangements for make-up work will be made on a case-by-case basis. This may include, but is not limited to, make-up assignments or an alternative exam schedule. Please note that it is at the discretion of the instructor to provide make-up work opportunities.
4. **Unexcused Absences**: Unexcused absences, i.e., those without prior notification or a valid reason, will not be taken lightly. **Please note that only two absences with prior notification will be accepted. Starting from the third absence, a penalty of a minimum of 10% will be applied to any assignments, labs, or exams missed due to an unexcused absence**. In certain cases, the student may not be allowed to make up for the missed coursework at all. Please note that it is at the discretion of the instructor to provide make-up work opportunities.

This attendance policy aims to foster a disciplined academic environment and ensure that each student can make the most of this learning opportunity. Your presence and active participation in every lecture are integral to your understanding of the course content, and any absence, excused or otherwise, will potentially have a significant impact on your academic performance.

Extra Credit and Grading Curves

This policy outlines the guidelines for the allocation of extra credit opportunities and the application of grading curves in this course. It is designed to foster consistent attendance, active class participation, and preserve the standards of our academic community as described in the Attendance and Expanded Academic Integrity policies.

1. **Eligibility for Extra Credit and Grading Curves**
   a. *Course Attendance*
      Regular attendance is essential for academic success. Students must attend **at least 70%** of the course lectures to qualify for **extra credit assignments** and to benefit from either any grading curves or any other extra credit opportunity implemented in this course. Attendance will be strictly monitored for each class session using an in-class attendance sheet or/and using other platforms (e.g., Canvas aPlus+ Attendance).
   b. *Academic Integrity Review*
      The instructor reserves the right to review or re-evaluate any submitted work on Canvas for plagiarism or other academic integrity concerns/academic integrity investigation at any time during the academic term, including assignments, projects, or assessments that have already been graded and returned to students. Students who are involved in an academic integrity investigation may not be eligible for extra credit or grading curves or any other extra credit opportunity implemented in this course. Students who are under investigation for a possible academic integrity violation may be temporarily ineligible for extra credit, grading curves, or other discretionary grade-enhancement opportunities offered in the course. If a student is found responsible for an academic integrity violation at any point during the semester, the student's eligibility for such opportunities may be revoked for the complete duration of the academic term. If an academic integrity violation is identified in previously graded work, the instructor may revise the original grade at any time during the semester, including assigning a grade of zero for the affected assessment, and may refer the matter to the Office of Student Conduct in accordance with university policies. This policy is intended to promote transparency, fairness, and accountability and to ensure that all students maintain ethical and responsible academic conduct throughout the semester.
   c. *Class Participation*
      Students who are actively involved in class discussions and actively participate in discussing the assigned reading materials and take part in class questions answer sessions may be eligible for extra credit or grading curves.
2. **Extra Credit**
   d. Extra credit assignments are intended to provide students with an opportunity to improve their grades by demonstrating a deeper understanding of the course material. However, these opportunities are contingent upon meeting the attendance and academic integrity requirements stated above.

3. **Grading Curves**
    e. When deemed appropriate, grading curves will be employed to more accurately reflect the student's comprehension of the course material. However, the advantage of grading curves will only be accessible to those students who meet the previously mentioned attendance and academic integrity requirements.

TEXAS TECH UNIVERSITY POLICIES / STATEMENTS

Academic Integrity and Plagiarism Statements

Academic integrity is taking responsibility for one's own class and/or course work, being individually accountable, and demonstrating intellectual honesty and ethical behavior. Academic integrity is a personal choice to abide by the standards of intellectual honesty and responsibility. Because education is a shared effort to achieve learning through the exchange of ideas, students, faculty, and staff have the collective responsibility to build mutual trust and respect. Ethical behavior and independent thought are essential for the highest level of academic achievement, which then must be measured. Academic achievement includes scholarship, teaching, and learning, all of which are shared endeavors. Grades are a device used to quantify the successful accumulation of knowledge through learning. Adhering to the standards of academic integrity ensures grades are earned honestly. Academic integrity is the foundation upon which students, faculty, and staff build their educational and professional careers. [Texas Tech University ("University") Quality Enhancement Plan, Academic Integrity Task Force, 2010].

Expanded Academic Integrity Policy

This course strongly advocates for an atmosphere of academic integrity and intellectual honesty. All students are encouraged to discuss ideas and problem-solving strategies with the Teaching Assistant, Instructor, and other students, but any form of academic dishonesty will not be tolerated. All submitted codes and assignments are subject to random checks for plagiarism.

1. **Code and Answers Sharing**
   Unless explicitly stated otherwise by the instructor, you are strictly prohibited from sharing, using, or looking at code or answers obtained from online sources, classmates, or friends. This also includes sharing, using, or viewing unauthorized solutions from previous iterations of the course or utilizing ChatGPT or other AI software to write code or devise pseudocode for you.
2. **Understanding Academic Dishonesty**
   It is your responsibility to educate yourself about what constitutes academic dishonesty. This extends beyond plagiarism and may include unauthorized collaboration, falsification, multiple submissions, and facilitating academic dishonesty.
3. **Clarifying Doubts**
   If you are unsure about the legitimacy of a specific action or behavior, it is essential to discuss it with the instructor before proceeding. This proactive approach can help avoid any unintentional violations of this policy.
4. **Penalties for Violation**
   Penalties for academic dishonesty will depend on the severity and frequency of the violations.
    - For a first-time violation on an assignment, the student will receive a score of '0' on the assignment in question.
    - Any student caught (1) violating this rule on an exam or project, (2) committing multiple violations, or (3) committing a severe enough violation as deemed by the professor will receive a grade of 'F' for the course, and their actions will be reported to the Office of Student Conduct.

AI Use Policy

To support your development and demonstration of the course learning outcomes, generative AI (such as ChatGPT, Claude, Grammarly, Copilot, or similar tools) use in this course is permitted only as specified for each assignment. Each graded activity will identify one of four AI Use Levels, from Level 0 through Level 3. Follow the stated level, including its permitted uses, prohibited uses, and documentation requirements. Basic grammar, spelling, and punctuation features, approved assistive technologies, and approved disability accommodations are

permitted unless they generate or substantially rewrite content or replace the skill being assessed. Ask before using an AI tool if you are unsure whether it is permitted. Use of generative AI outside these boundaries may be considered a violation of Texas Tech University's academic integrity policy. Please contact your instructor if you have any questions regarding this course policy. For additional information about the different levels of AI use, please visit the TTU Teaching, Learning, and Professional Development Center website: https://www.depts.ttu.edu/tlpdc/ai-syllabus-statements.php

ADA Statement

Any student who, because of a disability, may require special arrangements in order to meet the course requirements should contact the instructor as soon as possible to make any necessary arrangements. Students should present appropriate verification from Student Disability Services during the instructor's office hours. Any student who, because of a disability, may require special arrangements in order to meet the course requirements should contact the instructor as soon as possible to make any necessary arrangements. Students should present appropriate verification from Student Disability Services during the instructor's office hours. Please note: instructors are not allowed to provide classroom accommodations to a student until appropriate verification from Student Disability Services has been provided. For additional information, please contact Student Disability Services in Weeks Hall or call 806-742-2405.

Religious Holy Days

"Religious holy day" means a holy day observed by a religion whose places of worship are exempt from property taxation under Texas Tax Code §11.20. A student who intends to observe a religious holy day should make that intention known in writing to the instructor prior to the absence. A student who is absent from classes for the observance of a religious holy day shall be allowed to take an examination or complete an assignment scheduled for that day within a reasonable time after the absence. A student who is excused under section 2 may not be penalized for the absence; however, the instructor may respond appropriately if the student fails to complete the assignment satisfactorily.

Absence due to Officially Approved Trips

Students planning an approved absence must notify the instructor of their departure and return dates prior to the trip to obtain advance permission; please check these guidelines.

Accommodation for Pregnant Students

To support the academic success of pregnant and parenting students and students with pregnancy related conditions, the University offers reasonable modifications based on the student's particular needs. Any student who is pregnant or parenting a child up to age 18 or has conditions related to pregnancy may contact Alex Faris, the Texas Tech University designated Pregnancy and Parenting Liaison, to discuss support available through the University. The Liaison can be reached by emailing alfaris@ttu.edu. Should a student communicate with the instructor that they are pregnant or have a pregnancy related condition or may need additional resources related to pregnancy or parenting, the instructor will communicate that student's information to the Title IX Coordinator, who will work with the student and others, as needed, to ensure equal access to the University's education program or activity. For more information regarding supportive measures, please contact pregnancy & parenting liaison Alex Faris (alfaris@ttu.edu | 806.834.3420) or visit Pregnancy & Parenting. You can also submit a Pregnancy & Parenting Support Form to request assistance.

Late Arrival, Late Return, and Early Departure Policy

The Computer Science department strictly follows the official academic calendars and requires students who are enrolled in face-to-face sections to be on campus by the first class day of each semester and leave campus no earlier than the last day with scheduled course activities. The only exception we make is for incoming new international students who often need more time to obtain the necessary paperwork, including a study visa, and in such cases, we accommodate late arrival for up to the 12th class day of their first semester. No exceptions will be made for late return or early departure requests from current students in general. If it is because of an unforeseen and uncontrollable situation, a student needs early departure or late return; then the student must obtain in-advance

approval from the academic advisors and instructors of all enrolled courses for an excused absence of four (4) or fewer weekdays and an additional in-advance approved Extended Absence Verification* from the Office of the Dean of Students for an extended absence of five (5) or more weekdays. If a student has unexcused absences, then the student must take full responsibility for any missed classes, missed academic work, or any financial issues caused.

*Extended Absence Verification Request to be verified by Office of the Dean of Students: https://cm.maxient.com/reportingform.php?TexasTechUniv&layout_id=6

Food Insecurity

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students for support. The important thing is that all of the programs are now housed under the umbrella of The Raider Relief – Advocacy and Resource Center. It was developed to support student needs and connect them with essential resources necessary for academic success as a Texas Tech student. Furthermore, please notify the professor if you are comfortable in doing so. Raider Red's Food Pantry (located in Doak 117) supplies personal care items and a selection of nonperishable food to students. The Raider Relief Advocacy and Resource Center (RR- ARC) is a centralized hub of resources and support for students facing hardships with their basic needs. Through a comprehensive network of campus and community partnerships, we strive to alleviate the burden of financial, physical, and emotional hardships and promote the well-being and academic success of all students. Please check these outreach initiatives, Raider Relief Fund, Raider Red's Food Pantry, and Red to Black Peer Financial Coaching, and fill out our form to get connected: https://www.depts.ttu.edu/raiderrelief/.

Discrimination, Harassment, And Sexual Violence

Beginning January 1, 2020, Texas Education Code, Section 51.252 (formerly known as Senate Bill 212) requires all employees of Texas universities, including faculty, to report to the Title IX Office any information regarding incidents of sexual harassment, sexual assault, dating violence, or stalking that is disclosed to them. Texas law requires that all employees who witness or receive information about incidents of this type (including, but not limited to, written forms, applications, one-on-one conversations, class assignments, class discussions, or third-party reports) must report it to the Title IX Coordinator. Before talking with me or with any faculty or staff member about a Title IX-related incident, please remember that I will be required to report this information.

Texas Tech University is committed to providing and strengthening an educational, working, and living environment where students, faculty, staff, and visitors are free from gender and/or sex discrimination of any kind. Sexual assault, discrimination, harassment, and other Title IX violations are not tolerated by the University. Report any incidents to the Office for Student Rights & Resolution (806)-742-SAFE (7233) or file a report online at titleix.ttu.edu/students. Faculty and staff members at TTU are committed to connecting you to resources on campus. Some of these available resources are TTU Student Counseling Center, 806- 742-3674, https://www.depts.ttu.edu/scc/(Provides confidential support on campus.) TTU 24-hour Crisis Helpline, 806-742-5555, (Assists students who are experiencing a mental health or interpersonal violence crisis. If you call the helpline, you will speak with a mental health counselor.) Voice of Hope Lubbock Rape Crisis Center, 806-763-7273, voiceofhopelubbock.org (24-hour hotline that provides support for survivors of sexual violence.) The Risk, Intervention, Safety and Education (RISE) Office, 806-742-2110, https://www.depts.ttu.edu/rise/ (Provides a range of resources and support options focused on prevention education and student wellness.) Texas Tech Police Department, 806-742- 3931, http://www.depts.ttu.edu/ttpd/ (To report criminal activity that occurs on or near Texas Tech campus.). If you would like to have more information about reporting options and resources, I encourage you to visit the TTU website for more information or to contact the professional staff: https://www.depts.ttu.edu/titleix/students/Report_an_Incident.php.

Civility in the Classroom

Texas Tech University is a community of faculty, students, and staff that enjoys an expectation of cooperation, professionalism, and civility during the conduct of all forms of university business, including the conduct of

student-student and student-faculty interactions in and out of the classroom. Further, the classroom is a setting in which an exchange of ideas and creative thinking should be encouraged and where intellectual growth and development are fostered. Students who disrupt this classroom mission by rude, sarcastic, threatening, abusive, or obscene language and/or behavior will be subject to appropriate sanctions according to university policy. Likewise, faculty members are expected to maintain the highest standards of professionalism in all interactions with all constituents of the university.

Recovery Services

The Center for Students in Addiction Recovery offers students in recovery a nurturing and supportive community. The Center provides students in recovery with an abstinence-based program where students can flourish in recovery as they attain educational goals, including advanced degrees. The services provided through the CSAR increase the continuum of care for students in recovery, enhancing the quality of life for students in recovery at Texas Tech University. The CSAR supports students in recovery from alcohol, drugs, and behavioral addictions. By providing recovery support through relationships with staff, academic advising, scholarships/fellowships, recovery housing, study abroad opportunities, and more, students can flourish in recovery and in life.

Safety and Wellness:

The Texas Tech University (TTU) and Edward E. Whitacre Jr. College of Engineering are committed to the safety and wellness of our students by providing various services and resources.

Make sure you register with Tech Alert to get emergency notifications by phone call, text, or email. You are encouraged to review the Emergency Action Plans (EAPs) and watch the videos of Know What To Do In Emergency Events and Surviving an Active Shooter Event Training to be prepared for those emergency situations. Additionally, due to the nature of laboratory or design courses, it is mandatory for you to follow the university safety policies and any additional safety protocols required by the course instructor(s).

For your well-being, various services are available at Student Counseling Center and Student Health Services. The Student Wellness Center provides convenient walk-in services M-F from 8 am to 5 pm. Furthermore, the Texas Tech Crisis HelpLine (806-742-5555) provides 24/7/365 assistance for students experiencing a crisis or distress.

Emergency/Crisis Phone Number

TTU Police (UPD) Emergency	911
TTU Police (UPD) Non-Emergency	806.742.3931
TTU Emergency Maintenance	806.742.4677
TTU EHS (M-F, 8 am – 5 pm)	806.742.3876
SafeRide	806.742.7433
TTU Crisis HelpLine	806.742.5555
Student Wellness Center (From Urgent Care to a Full-Service Pharmacy on site)	806.742.2848
Title IX Reporting	806.742.7233
The Dean of Students	806.742.2984

Course Resources/ Tools

***Important Note*:** The topics (below table), the order in which they are presented, and the information presented in this complete syllabus are subject to change, expansion, and contraction or stasis during the semester at the instructor's discretion because of scheduling issues, developments in the discipline, or other contingencies. The copy of syllabus on RaiderCanvas takes precedence in case of conflict between different versions of syllabus. This schedule is also tentative and subject to change.

Week of...	Monday	Wednesday	Friday
08/24/2026	Syllabus Overview and Course Introduction	Software Engineering, Community-Engaged Learning	Community Partner Context; CeL Project Overview; Team-Interest Form
08/31/2026	Git and GitHub (Branches, Pull Requests, Issues, and Code Review)	Git and GitHub (Branches, Pull Requests, Issues, and Code Review)	Partner Orientation; Team Formation; Stakeholder Mapping
09/07/2026	School Holiday	JavaScript, Software Processes and Software Development Life-Cycle Models	JavaScript, Agile Development: Scrum/Kanban, Estimation and Risk
09/14/2026	React, Requirements Elicitation	React, Stakeholder Interviews, Observation, and Current-Workflow Analysis	React, Stakeholder Engagement; Problem Discovery
09/21/2026	React, Requirements Analysis and Problem Framing	Engineering Job Fair	React, Ethics, Privacy, Needs-Assessment, User Stories, Use Cases
09/28/2026	Software Requirements Specification	Requirements Validation, Traceability, and Quality	Change Management; Partner Requirements Review; Software Requirements Specification (SRS) Submission
10/05/2026	User Experience and Interface Design	Prototyping and Accessibility	Data and Interface Modeling; Partner Co-Design/Prototype Review
10/12/2026	Software Architecture and System Modeling	UML, Design Patterns, and Architectural Trade-Offs	Architecture/Prototype Package
10/19/2026	Architecture Review /Guest Lecture	Code Review and Implementation Practices	Code Review and Implementation Practices
10/26/2026	APIs and Software Integration	Databases and Data Management	Deployment Environments, Docker/Cloud Concepts
11/02/2026	Verification and Validation	Unit and Integration Testing	System, Acceptance, and Regression Testing; Test Planning
11/09/2026	Secure Software Design and Risk Analysis	Privacy, Accessibility, and Quality Assurance	Partner/User Validation; Defect Triage; Security and Accessibility Evidence
11/16/2026	Software Maintenance, Reliability, Observability, and Release Planning	Software Maintenance, Reliability, Observability, and Release Planning	Release-Candidate Review; Sprint and Project-Management Evidence
11/23/2026	Documentation, Training, Deployment, Transition, and Sustainability	School Holiday	School Holiday
11/30/2026	Final Project Demonstrations	Final Project Demonstrations	X